MIL-S-19500/200A
19 April 1968
SUPERSEDING
MIL-S-19500/200
6 June 1961
(See 6.2)

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, DIODE, GERMANIUM

TYPE 1N270

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for a germanium semiconductor diode and is in accordance with MIL-S-19500 except as otherwise specified herein.

1.2 Physical dimensions. See figure 1 (DO-7).

1.3 Ratings and characteristics.

v_r	V_{RM}(wkg)	I_o 1/	i_f	i_f(surge) 1/120 sec
v(pk)	v(pk)	mAdc	ma	a
100	80	60	325	1

1/ Derate at 0.923 mAdc/°C above T_A = 25°C.

OPERATING AMBIENT TEMPERATURE: -65° to +75°C.

STORAGE TEMPERATURE: -65° to +90°C.

BAROMETRIC PRESSURE, REDUCED: 8 mm Hg.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARDS

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 5961

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3. REQUIREMENTS

3.1 General. Requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations and symbols. The abbreviations and symbols used herein are defined in MIL-S-19500.

3.3 Design, construction, and physical dimensions. Diodes shall be of the design, construction, and physical dimensions shown in figure 1.

3.4 Performance characteristics. Performance characteristics shall be as specified in tables I, II, and III.

3.5 Marking. The following marking specified in MIL-S-19500 may be omitted at the option of the manufacturer:

(a) Manufacturer's identification.
(b) Country of origin.

3.5.1 Polarity. The polarity shall be indicated with a contrasting color band to denote the cathode end.

3.6 Hermetic seal. As part of the manufacturing process, all diodes procured under this specification shall be leak tested as follows: The diodes shall be immersed for a minimum of 15 minutes in a pressure tank with a suitable dye such as fluorescein or Rhodamine B penetrant under 100 psig. After washing the diodes, each diode shall be examined under a 20X magnification where Rhodamine B has been used and visually under ultraviolet light where fluorescein dye has been used. Any device exhibiting dye penetration into the seal of the body shall be removed from the lot.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall consist of the examinations and tests specified in tables I, II, and III.

4.3 Quality conformance inspection. Quality conformance inspection shall consist of the examination and tests specified in Groups A, B, and C.

4.3.1 Group A inspection. Group A inspection shall consist of the examinations and tests specified in table I.

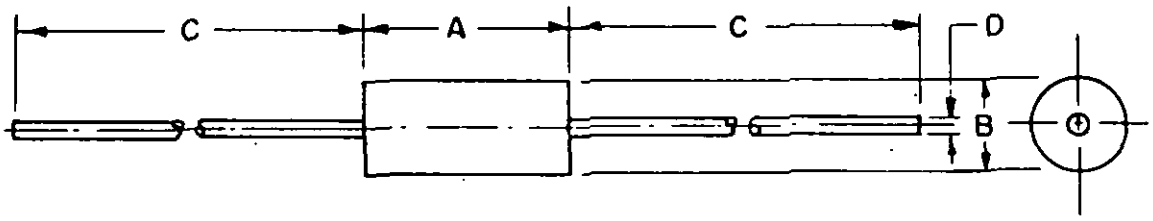
4.3.2 Group B inspection. Group B inspection shall consist of the examinations and tests specified in table II.

4.3.3 Group C inspection. Group C inspection shall consist of the examinations and tests specified in table III. This inspection shall be conducted on the initial lot, and thereafter every six months during production.

4.4 Methods of examination and test. Methods of examination and test shall be as specified in tables I, II, and III, and as follows:

4.4.1 Steady-state operation life. This test shall be conducted with a half-sine wave of the specified peak voltage impressed across the diode in the reverse direction followed by a half-sine wave form of the specified average rectified current. The forward conduction angle of the rectified current shall not be greater than 180° nor less than 130°, and the power shall be equal to or greater than that of a half-sine wave.

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Ltr	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
A	.230	.300	5.84	7.62	
B	.092	.130	2.34	3.30	4
C	1.000	1.500	25.40	38.10	3
D	.018	.022	.46	.56	1,2&3

NOTES:
1. The specified lead diameter applies in the zone between .050 (1.27 mm) and 1.000 (25.40 mm) from the diode body. Outside of this zone the lead diameter is not controlled.
2. Gold plated or tinned leads may be supplied providing units conform to subgroups 2 and 4 of Group B inspection and subgroup 1 of Group C inspection.
3. Both leads shall be within the specified dimension.
4. The minimum body diameter shall be maintained over .15 (.38 mm) of body length.
5. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

FIGURE 1. Semiconductor device, diode, type 1N270(DO-7).

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4.4.2 Instability criteria. The diode shall be held firmly by the body and vibrated for a minimum of 10 seconds so as to apply a peak acceleration load of 10 G minimum at approximately 60 Hz, while the reverse current trace is being observed on an oscilloscope with the diode reverse voltage being swept out to a value at least as great as the highest reverse test voltage. There shall be no break in the oscilloscopic observation of the trace showing noise generation due to poor junction contact, extraneous material, or other defect within the device.

4.4.3 Time limit for end points. End point tests for qualification and quality conformance inspection shall be completed within 96 hours after completion of the last test in the subgroup.

TABLE I. Group A inspection

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		
	Method	Details			Min	Max	Unit
Subgroup 1			10				
Visual and mechanical examination	2071			---	---	---	---
Subgroup 2			5				
Forward voltage	4011	I_F = 200 mAdc		V_F	---	1.0	Vdc
Reverse current	4016	DC method; V_R = 80 Vdc		I_R	---	100	µAdc
Peak reverse current at peak reverse voltage	4016	AC method; v_r = 100 v(pk)		i_r	---	1.0	ma(pk)
Reverse current	4016	DC method; T = 75°C; V_R = 10 Vdc		I_R	---	75	µAdc

TABLE II. Group B inspection

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		
	Method	Details			Min	Max	Unit
Subgroup 1			10				
Physical dimensions	2066	(See figure 1)		---	---	---	---
Subgroup 2			10				
Solderability	2026	Immersion depth to within 0.10 inch of body		---	---	---	---
Thermal shock (temperature cycling)	1051	Test condition A		---	---	---	---
Thermal shock (glass strain)	1056	Test condition A		---	---	---	---
Terminal strength: Tension	2036	Test condition A; weight 4 lb; t = 15 ±3 sec		---	---	---	---
Moisture resistance	1021			---	---	---	---

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TABLE II. Group B inspection - Continued

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		
	Method	Details			Min	Max	Unit
Subgroup 2 - Continued			10				
End points: (See 4.4.3)							
Forward voltage	4011	I_F = 200 mAdc		V_F	---	1.0	Vdc
Reverse current	4016	DC method; V_R = 80 Vdc		I_R	---	100	µAdc
Subgroup 3			10				
Shock	2016	Nonoperating; 1,500 G; t = 0.5 msec; 5 blows each orientation X_1, Y_1, and Y_2		---	---	---	---
Instability (see 4.4.2)	---			---	---	---	---
Vibration fatigue	2046	Nonoperating; X_1, Y_1, and Y_2 orientations		---	---	---	---
Vibration, variable frequency	2056	Nonoperating; X_1 and Y_1 orientations		---	---	---	---
Constant acceleration	2006	Nonoperating; 20,000 G; X_1, Y_1, and Y_2 orientations		---	---	---	---
End points: (Same as subgroup 2)							
Subgroup 4			10				
Terminal strength: Lead fatigue	2036	Test condition E		---	---	---	---
End points: (Same as subgroup 2)							
Subgroup 5			10				
Surge current	4066	I_O = 60 mAdc; i_f(surge) = 1.0 a; t_p = 1/120 sec, 10 surges at 1/minute		---	---	---	---
End points: (Same as subgroup 2)							
Subgroup 6			λ = 5				
High-temperature life (nonoperating)	1031	T_A = 90°C		---	---	---	---
End points: (See 4.4.3)							
Forward voltage	4011	I_F = 200 mAdc		V_F	---	1.0	Vdc
Reverse current	4016	DC method; V_R = 80 Vdc		I_R	---	150	µAdc

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TABLE II. Group B inspection - Continued

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		
	Method	Details			Min	Max	Unit
Subgroup 7			$\lambda = 5$				
Steady-state operation life (see 4.4.1)	1026	f = 60 Hz; I_O = 60 mAdc; v_r = 80 v(pk)		---	---	---	---
End points: (Same as subgroup 6)							

TABLE III. Group C inspection

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		
	Method	Details			Min	Max	Unit
Subgroup 1			10				
Barometric pressure, reduced (altitude operation)	1001	8 mm Hg; t = 60 sec		---	---	---	---
Measurement during test:							
Reverse current	4016	DC method; V_R = 80		I_R	---	150	μAdc
Subgroup 2			10				
Salt atmosphere (corrosion)	1041			---	---	---	---

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery. Preparation for delivery shall be in accordance with MIL-S-19500.

6. NOTES

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Changes from the previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - EL
Navy - EC
Air Force - 11

Review activities:
Army - EL, MU
Navy - SH
Air Force - 11, 17, 85
DSA - ES

User activities:
Army - EL, MI SM
Navy - AS, OS, MC, CG
Air Force - 19

Preparing activity:
Navy - EC

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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